Student Record and Academic Management System

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SUBJECT: Python

Division: A

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Problem Statement

Managing student academic records manually using registers, notebooks, spreadsheets, or separate files can be time-consuming and difficult. Searching for student information, updating records, maintaining marks, and calculating total marks, average marks, percentages, and grades may require significant effort and can lead to errors.

As the number of students increases, managing and organizing records becomes more complicated. Records may also be misplaced, duplicated, or incorrectly updated, making it difficult to maintain accurate information and monitor student performance.

The proposed Student Record and Academic Management System aims to provide a simple and organized solution for managing student details and academic records. The system will allow users to add, search, update, and view student information, maintain subject-wise marks, and automatically calculate averages and grades.

The main purpose of this project is to understand the problems of manual record management, identify system requirements, and design an efficient software-based solution before actual implementation.

Key Requirements

In short: The system will provide basic student record management, searching, information display, and automatic academic performance calculations.

Feature	Purpose
Add Student	Add a new student's personal information and subject marks.
Search Student	Find a student using their Roll Number or PRN.
View Student Details	Display complete student information, including academic marks.
Calculate Average Marks	Calculate the student's average marks from all subject marks.
Calculate Grade	Determine the student's grade based on their average marks.
Exit	Safely exit the system.

Inputs and Outputs

Note: The Student Record and Academic Management System takes student information such as Roll Number, Name, PRN, Class, Division, subjects, and marks as input. The system stores and searches student records using Roll Number or PRN. It calculates the total marks and average marks and determines the student's grade based on predefined grading rules. The system displays student details, successful record confirmation, search results, or "Student Not Found" when no matching record exists.

Input	Process	Output
Student record	Store student information	Student details / Confirmation
Roll Number / PRN	Search the student record	Student details / Student Not Found
Subject marks	Calculate total marks	Total Marks
Subject marks	Calculate average marks	Average Marks
Average marks	Apply grading rules	Grade

Algorithm

Algorithm 1 – Add Student Record

1.Start.

2.Enter Roll Number, Name, PRN, Class, and Division.

3.Enter marks for all required subjects.

4.Store the student details and marks.

5.Display "Student Added Successfully".

6.Stop.

Algorithm 2 – Search Student Record

1.Start.

2.Enter the student's Roll Number.

3.Search the student records.

4.Check whether the Roll Number exists.

4.If found, display the student's details.

5.Otherwise, display "Student Not Found".

6.Stop.

Algorithm 3 – Calculate Academic Result

1.Start.

2.Enter marks for all required subjects.

3.Calculate Total Marks.

4.Calculate Average Marks.

5.Compare the average with the grading rules.

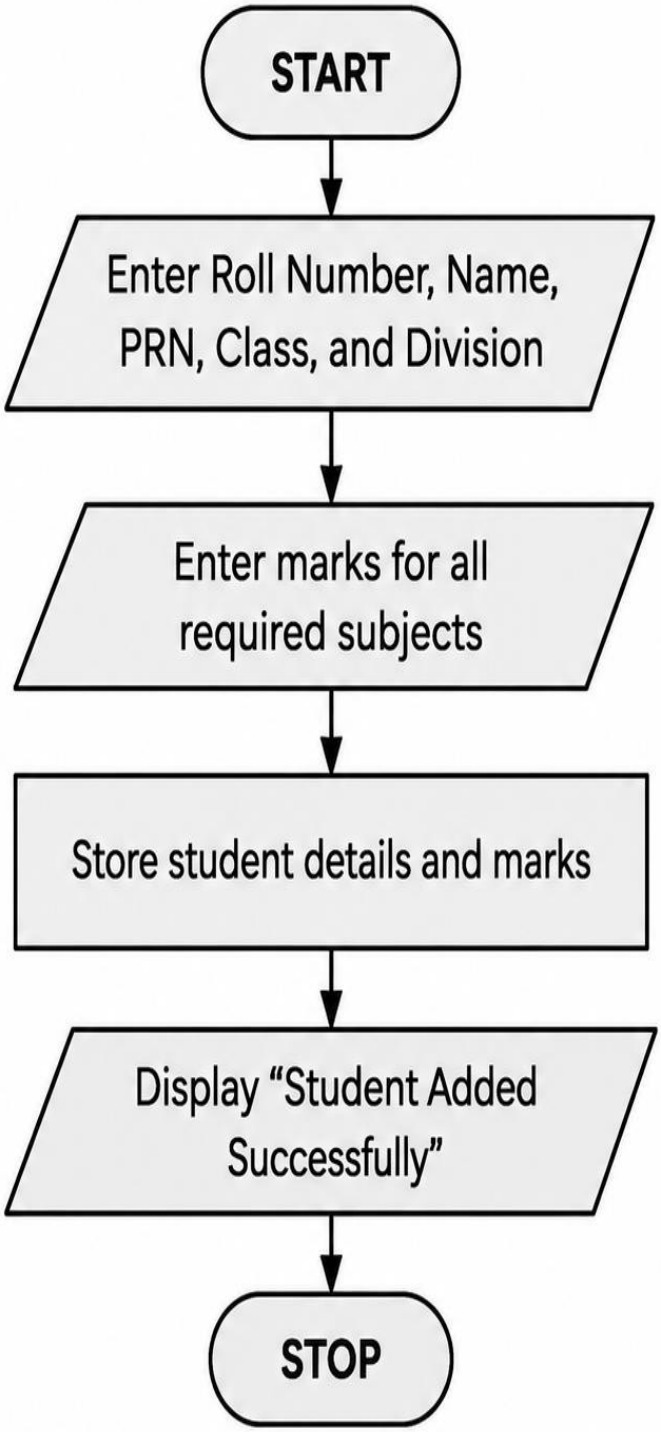
6.Assign the appropriate Grade.

7.Display Total Marks, Average Marks, and Grade.

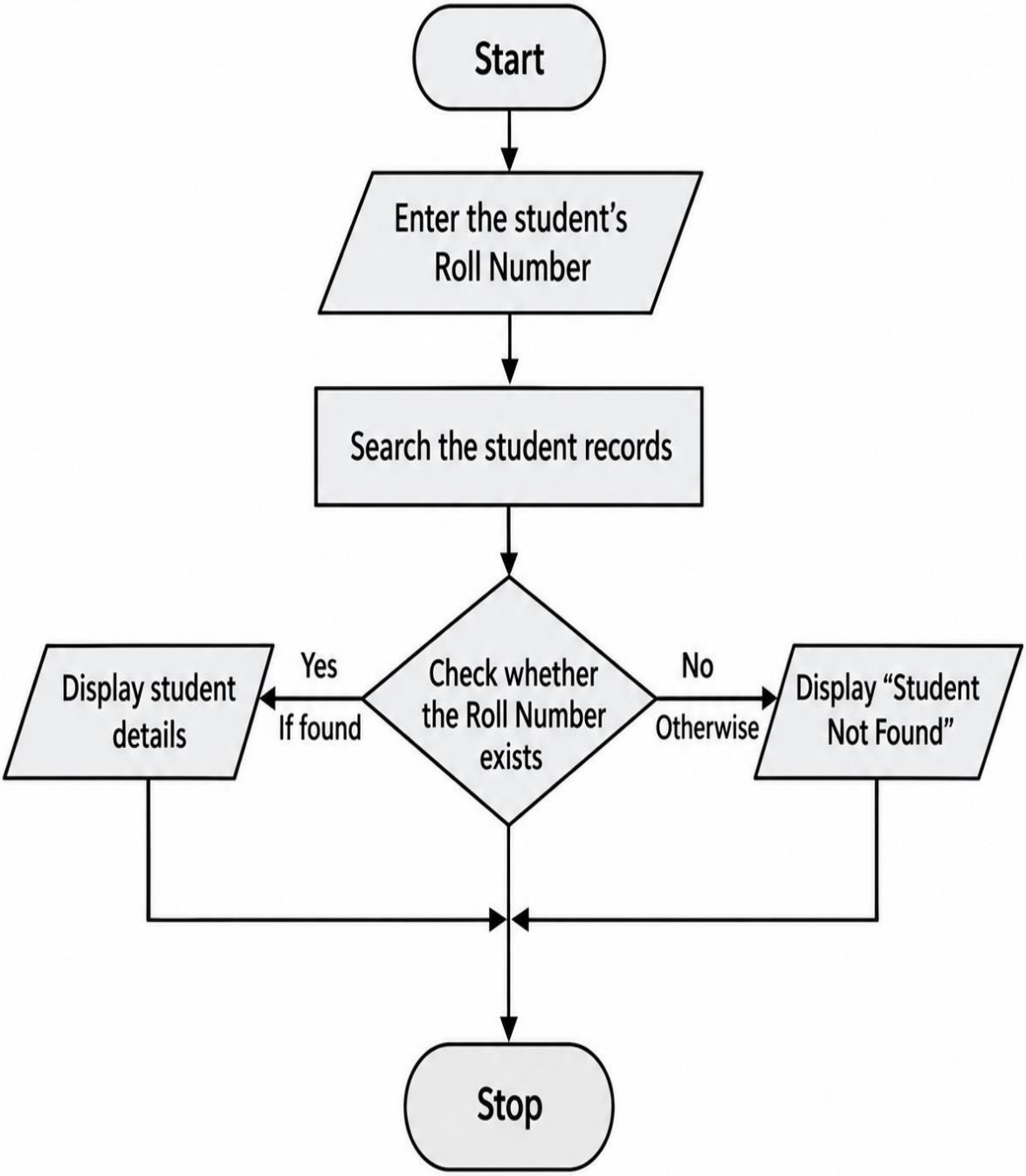
8.Stop.

Flowchart Collection

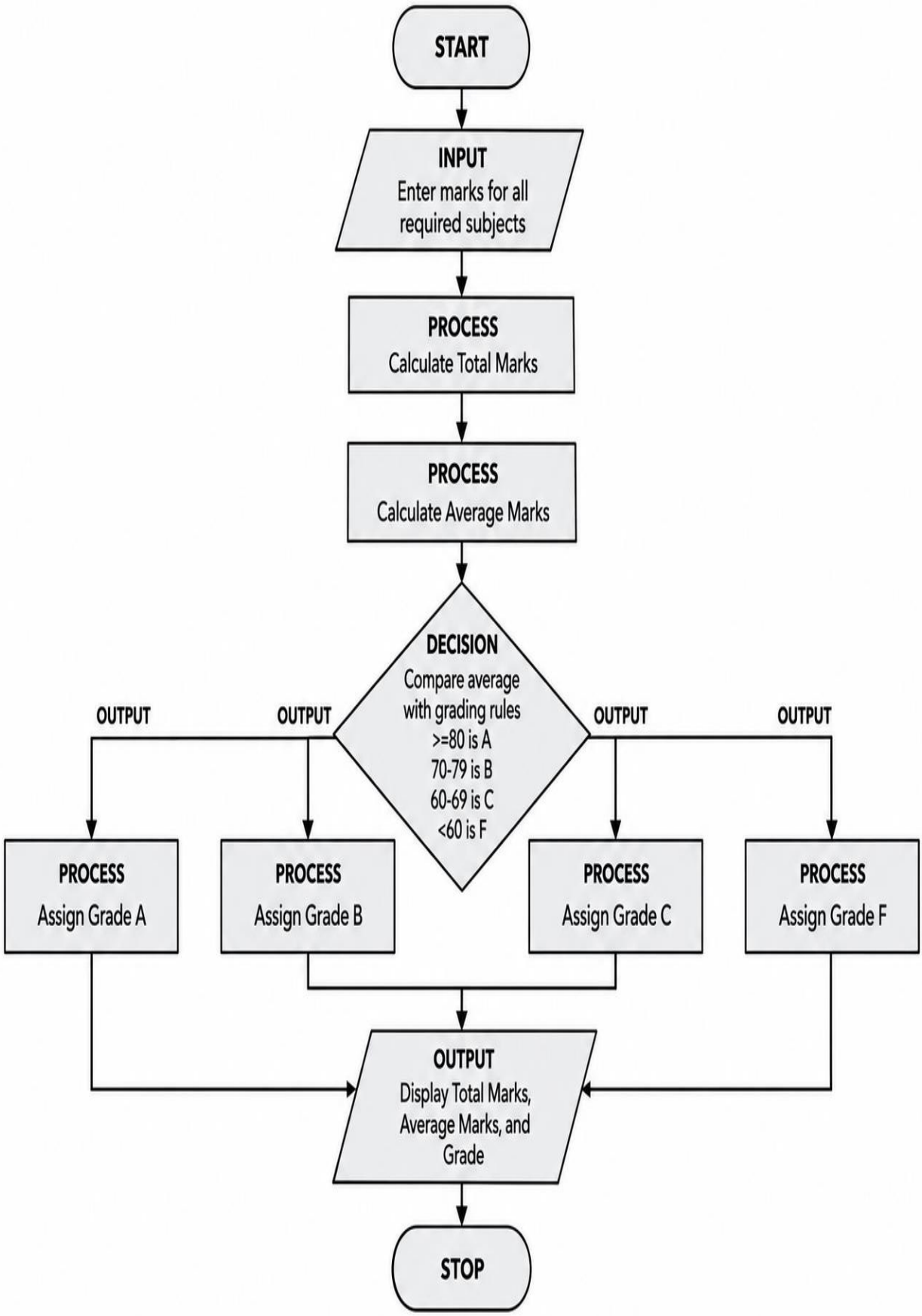
Algorithm 1 – Add Student Record



Algorithm 2 – Search Student Record



Algorithm 3 – Calculate Academic Result



Pseudocode Document

1.Pseudocode – Add Student Record

```
BEGIN

    INPUT Roll_Number

    SEARCH student records using Roll_Number

    IF Roll_Number is found THEN
        DISPLAY student details
    ELSE
        DISPLAY "Student Not Found"
    END IF

END
```

2.Pseudocode – Search Student Record

```
BEGIN

    INPUT Roll_Number

    SEARCH student records using Roll_Number

    IF Roll_Number is found THEN
        DISPLAY student details
    ELSE
        DISPLAY "Student Not Found"
    END IF

END
```

3.Pseudocode – Calculate Academic Result

```
BEGIN

    INPUT marks for all required subjects

    CALCULATE Total_Marks
    CALCULATE Average_Marks

    IF Average_Marks >= 80 THEN
        Grade = "A"
    ELSE IF Average_Marks >= 70 THEN
        Grade = "B"
    ELSE IF Average_Marks >= 60 THEN
        Grade = "C"
    ELSE
        Grade = "F"
    END IF

    DISPLAY Total_Marks
    DISPLAY Average_Marks
    DISPLAY Grade

END
```

Short Explanation of Proposed Solution

The proposed Student Record and Academic Management System is designed to provide a simple and organized way to manage student information and academic records. It will store important details such as Roll Number, Name, PRN, Class, Division, and subject-wise marks in one system.

The system will provide different operations such as Add Student, Search Student, View Student Details, Calculate Average Marks, and Calculate Grade. When adding a student, the user will enter the required personal and academic information, which will then be stored for future use.

The Search Student feature will allow the user to find a particular student's record using the Roll Number. If the record is available, the system will display the student's details; otherwise, it will show "Student Not Found."

The system will also process the marks entered for each subject. It will calculate the total marks and average marks and determine the student's grade according to the approved grading criteria. This reduces manual calculations and helps avoid common calculation errors.

Overall, the proposed system will make student record management faster, easier, organized, and more reliable. It can also provide a foundation for adding more features in the future, such as updating records, deleting records, generating reports, and storing data in a database.